Lecture Notes

Lecture 3

Linear Maps: Kernel, Image, Rank–Nullity

We have the objects of linear algebra—vector spaces—and their coordinate systems—bases. Now we study the *maps between* vector spaces that respect this structure. A *linear map* is a function that commutes with addition and scalar multiplication; equivalently, it preserves linear combinations, so it preserves superposition.

This is also why linear maps are the right notion of a "transformation" in physics. In quantum mechanics linear operators do three different jobs. One kind of operator encodes a measurable quantity. A second describes how a closed system evolves in time. A third describes a symmetry. Measurement itself does something that is not a linear operation: it selects the part of a state belonging to one possible outcome and replaces the state by that part. We will define all of this in Lectures 7–10, and nothing in the present lecture depends on it.

Two subspaces measure how a linear map behaves: its *kernel* (what it sends to zero) and its *image* (what it can produce). The central result of the lecture, the rank–nullity theorem, ties their dimensions together in a single equation $\dim V = \dim \ker T + \dim \operatorname{im} T$, and from it we will read off when linear systems are solvable and when operators are invertible.

Learning objectives

By the end of this lecture you will be able to:

- Check that a map is linear.
- Build a linear map from its values on a basis.
- Compute the kernel and image and read off injectivity and surjectivity.
- Apply the rank–nullity theorem and its dimension obstructions.
- Recognise the quotient space and First Isomorphism Theorem (*self-study*).

1 Linear maps and their examples

Definition 3.1 (Linear map). Let V, W be vector spaces over the same field $\mathbb{F}$. A function $T : V \to W$ is a *linear map* (or *linear transformation*) if

$$T(u+v) = Tu + Tv \quad \text{and} \quad T(av) = a\,Tv \qquad \text{for all } u, v \in V,\ a \in \mathbb{F}.$$

The set of all linear maps $V \to W$ is written $\mathcal{L}(V, W)$. A linear map from V to itself is an *operator*, and we write $\mathcal{L}(V)$ for $\mathcal{L}(V, V)$.

Applying the two rules repeatedly, a linear map sends 0 to 0 (take $a = 0$) and preserves every linear combination:

$$T(a_1 v_1 + \cdots + a_m v_m) = a_1\,Tv_1 + \cdots + a_m\,Tv_m. \tag{3.1}$$

This single identity is the reason linear maps are ubiquitous.

Example 3.2 (Linear maps). Each of the maps listed below is linear; both rules of the definition are checked directly.

(1) The *zero and identity maps*, $0 : V \to W$, $0\,v \mapsto 0$ and $I : V \to V$, $Iv = v$.

(2) *Matrix maps*: for $A \in M_{m\times n}(\mathbb{F})$, the map $x \mapsto Ax$ from $\mathbb{F}^n$ to $\mathbb{F}^m$. In Lecture 4 we will see that, in coordinates, *every* linear map looks like this.
(3) *Differentiation* $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Dp = p'$, and more generally on spaces of differentiable functions.
(4) *Multiplication* $M\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $(Mp)(t) = t\,p(t)$.
(5) The *backward shift* on $\mathbb{F}^\infty$, $T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$.
(6) *Rotation* of $\mathbb{R}^2$ by an angle θ, and the *evaluation* functional $\mathcal{P}(\mathbb{F}) \to \mathbb{F}$, $p \mapsto p(c)$. ◁

Example 3.3 (Checking linearity).

(1) The map $S(x, y) = (2x - y,\ x)$ on $\mathbb{R}^2$ is linear: each output coordinate is a linear combination of x, y with no constant term.
(2) The map $T(x, y) = (x + 1,\ y)$ is *not* linear—it fails $T0 = 0$, since $T(0, 0) = (1, 0) \neq 0$. A constant term in any coordinate destroys linearity. ◁

Example 3.4 (The key rule). Differentiation is linear because of a single identity

$$(af + bg)' = af' + bg'.$$

The same pattern—the rule is applied to each piece of the sum separately—holds for the backward shift, integration, multiplication by a fixed function, and matrix multiplication. That is why every map in Example 3.2 is linear. Linearity is usually easy to check. The harder question is what a given linear map *does*. The rest of the lecture is devoted to it. ◁

2 A linear map is determined by a basis

A linear map cannot be chosen freely at every point: (3.1) ties its values together. In fact its values on a single basis fix it completely, and those values may be assigned arbitrarily.

> **Theorem 3.5 (Maps are determined by a basis).** Let $v_1, \dots, v_n$ be a basis of V and let $w_1, \dots, w_n$ be *any* vectors in W. Then there is a unique linear map $T\colon V \to W$ with $Tv_k = w_k$ for each k.

Proof. Existence. Each $v \in V$ has a unique expression $v = \sum_k a_k v_k$ (Lecture 2); define $Tv = \sum_k a_k w_k$. Linearity follows because coordinates add and scale: if $v = \sum a_k v_k$ and $u = \sum b_k v_k$ then $v + u = \sum (a_k + b_k) v_k$, so $T(v + u) = \sum (a_k + b_k) w_k = Tv + Tu$, and similarly $T(cv) = c\,Tv$. Clearly $Tv_k = w_k$.

Uniqueness. If S is linear with $Sv_k = w_k$, then for any $v = \sum a_k v_k$, (3.1) gives $Sv = \sum a_k w_k = Tv$. So $S = T$. ∎

> **Corollary 3.6 (Agreement on a basis).** If two linear maps $S, T \in \mathcal{L}(V, W)$ satisfy $Sv_k = Tv_k$ on a basis $v_1, \dots, v_n$ of V, then $S = T$.

Proof. Both are linear maps with the same values on the basis, so both equal the unique map of Theorem 3.5. ∎

Example 3.7 (Building a map from its values on a basis). Define $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ by its action on the basis $(1, 1), (1, -1)$:

$$T(1, 1) = (2, 0), \qquad T(1, -1) = (0, 2).$$

To evaluate $T(x, y)$, we first find the coordinates of the vector (x, y) in this basis. We look for scalars a, b such that $(x, y) = a(1, 1) + b(1, -1)$. We write the basis vectors into the columns

of a matrix and the vector (x, y) on the right, and reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & x \\ 1 & -1 & y \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} \textcircled{1} & 0 & \frac{x+y}{2} \\ 0 & \textcircled{1} & \frac{x-y}{2} \end{array}\right] \implies a = \frac{x+y}{2}, \quad b = \frac{x-y}{2}.$$

Then

$$T(x, y) = a\,(2, 0) + b\,(0, 2) = (2a,\ 2b) = (x + y,\ x - y).$$

Theorem 3.5 guarantees this is the one and only linear map with the prescribed values. ◁

3 The algebra of linear maps

Linear maps can be added, scaled, and composed.

Definition 3.8 (Operations on linear maps). For $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, define $S + T$ and aS pointwise:

$$(S + T)v = Sv + Tv \quad \text{and} \quad (aS)v = a(Sv).$$

For $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, the *product* (composition) $ST : U \to W$ is $(ST)u = S(Tu)$.

Addition and scaling are defined for maps between the same pair of spaces. The product ST is defined only when the codomain of T is the domain of S. So the proposition states, for every map, between which spaces it acts.

Proposition 3.9 (The algebra of linear maps).

(1) With these operations, $\mathcal{L}(V, W)$ is a vector space.

(2) The product of linear maps $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ is linear too: $ST \in \mathcal{L}(U, W)$.

(3) The product is associative, distributes over addition, and has the identity maps as units. That is, for all $T, T_1, T_2 \in \mathcal{L}(U, V)$, $S, S_1, S_2 \in \mathcal{L}(V, W)$ and $R \in \mathcal{L}(W, X)$,

$$(RS)T = R(ST), \qquad S(T_1 + T_2) = ST_1 + ST_2, \qquad (S_1 + S_2)T = S_1T + S_2T,$$

and, for the identity maps I_V and I_W of V and W,

$$I_W S = SI_V = S.$$

Proof. (1) $\mathcal{L}(V, W)$ is a subspace of the space W^V of all functions $V \to W$: the zero map is linear, and if S, T are linear and $a \in \mathbb{F}$ then so are $S + T$ and aS, since, for example,

$$(S + T)(u + v) = S(u + v) + T(u + v) = Su + Sv + Tu + Tv = (S + T)u + (S + T)v.$$

(2) Apply the linearity of T, then the linearity of S:

$$(ST)(u + u') = S(Tu + Tu') = (ST)u + (ST)u' \quad \text{and} \quad (ST)(au) = S(a\,Tu) = a\,(ST)u.$$

(3) Each identity is checked by applying both sides to an arbitrary vector $u \in U$. For example, by the linearity of S,

$$\big(S(T_1 + T_2)\big)u = S(T_1u + T_2u) = S(T_1u) + S(T_2u) = (ST_1 + ST_2)u.$$

The remaining identities are checked in the same way. ■

For operators, that is when $U = V = W = X$, all these products are defined. So operators on V can be added, scaled and multiplied with one another, and the result is again an element of $\mathcal{L}(V)$. Products of operators, however, **need not commute**—and the failure to commute carries physical content.

Example 3.10 (A commutator: the canonical commutation relation). On $\mathcal{P}(\mathbb{R})$ let D be differentiation and M multiplication by t. For any p,

$$(DM)p = \big(t\,p(t)\big)' = p(t) + t\,p'(t), \qquad (MD)p = t\,p'(t),$$

so $(DM - MD)p = p$ for every p; that is, the *commutator* $[D, M] := DM - MD$ equals the identity:

$$[D, M] = I. \tag{3.2}$$

In quantum mechanics the position and momentum operators obey the analogous relation $[\hat{x}, \hat{p}] = i\hbar I$. Non-commuting operators are more the rule than the exception. Equation (3.2) is the first example of this in the course. ◁

Example 3.11 (Concrete non-commuting operators). The phenomenon is visible already with 2×2 matrices. Let $S, T \in \mathcal{L}(\mathbb{R}^2)$ be the matrix maps with

$$S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$ST = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad TS = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad [S, T] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0.$$

In physics S and T are the raising and lowering operators of a two-level system, for example an electron spin or a qubit. ◁

Remark 3.12. Once bases of finite-dimensional V and W are fixed, addition, scaling, and composition of linear maps become exactly addition, scaling, and multiplication of their matrices. This is the subject of Lecture 4. One can show that $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$, but we defer the proof: it is simpler in matrix language.

4 Kernel and injectivity

Definition 3.13 (Kernel (null space)). The *kernel* (or *null space*) of $T \in \mathcal{L}(V, W)$ is

$$\ker T = \{v \in V : Tv = 0\} \subseteq V.$$

In other words, the kernel consists of all the vectors that T sends to the zero vector.

Proposition 3.14 (The kernel is a subspace). $\ker T$ is a subspace of V.

Proof. $T0 = 0$, so $0 \in \ker T$. If $u, v \in \ker T$ and $a \in \mathbb{F}$ then $T(u + v) = Tu + Tv = 0$ and $T(av) = a\,Tv = 0$, so $u + v$ and av lie in $\ker T$. The subspace test is met. ∎

Example 3.15 (Kernels).

(1) The kernel of differentiation on $\mathcal{P}(\mathbb{R})$, $p \mapsto p'$, is the constants—exactly the polynomials with zero derivative.

(2) The kernel of multiplication by t on $\mathcal{P}(\mathbb{R})$, $p \mapsto t\,p$, is $\{0\}$ (here 0 is the zero polynomial).

(3) The kernel of the backward shift on $\mathbb{F}^\infty$, $(x_1, x_2, \dots) \mapsto (x_2, x_3, \dots)$, is $\{(a, 0, 0, \dots) : a \in \mathbb{F}\}$.

(4) The kernel of a matrix map on $\mathbb{F}^n$, $x \mapsto Ax$, is the solution set of the homogeneous system $Ax = 0$—the null space you met when row-reducing A. ◁

Example 3.16 (Computing a kernel). Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ be given by $Tx = Ax$ with $A = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{smallmatrix}\right)$. To find the kernel $\ker T$ we reduce the matrix A (there is no need to write the right-hand side, since it stays zero throughout):

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 2 & 0 \\ 0 & 0 & \textcircled{1} \end{bmatrix} \Longrightarrow x_1 = -2x_2, \quad x_3 = 0.$$

The pivots are in the first and third columns, so the only free variable is x_2. Setting $x_2 = 1$ reads off the basis vector $(-2, 1, 0)$, so

$$\ker T = \operatorname{span}\big((-2, 1, 0)\big), \qquad \dim \ker T = 1.$$

We return to this map in Example 3.22 to find its image. ◁

Definition 3.17 (Injective). $T\colon V \to W$ is *injective* (one-to-one) if $Tu = Tv$ implies $u = v$.

Theorem 3.18 (Injectivity is a kernel condition). $T \in \mathcal{L}(V, W)$ is injective if and only if $\ker T = \{0\}$.

Proof. If T is injective and $v \in \ker T$, then $Tv = 0 = T0$ forces $v = 0$, so $\ker T = \{0\}$. Conversely, if $\ker T = \{0\}$ and $Tu = Tv$, then $T(u - v) = Tu - Tv = 0$, so $u - v \in \ker T = \{0\}$, giving $u = v$. Thus T is injective. ∎

5 Image and surjectivity

Definition 3.19 (Image; rank). The *image* of $T \in \mathcal{L}(V, W)$ is

$$\operatorname{im} T = \{Tv : v \in V\} \subseteq W.$$

When the image $\operatorname{im} T$ is finite-dimensional, its dimension $\dim \operatorname{im} T$ is called the *rank* of T. This is always the case when V is finite-dimensional, since the images of a basis of V span $\operatorname{im} T$. The map T is *surjective* (onto) if $\operatorname{im} T = W$.

Proposition 3.20 (The image is a subspace). $\operatorname{im} T$ is a subspace of W.

Proof. $0 = T0 \in \operatorname{im} T$. If $w_1 = Tv_1$ and $w_2 = Tv_2$ then $w_1 + w_2 = T(v_1 + v_2) \in \operatorname{im} T$, and $aw_1 = T(av_1) \in \operatorname{im} T$. The subspace test holds. ∎

Example 3.21 (Image and codomain). The image of a matrix map $x \mapsto Ax$ is the span of the columns of A—the column space. Differentiation $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is surjective, since every polynomial has an antiderivative. But surjectivity depends on the chosen codomain: $D\colon \mathcal{P}_5(\mathbb{R}) \to \mathcal{P}_5(\mathbb{R})$ is *not* surjective (t^5 is not a derivative of anything in $\mathcal{P}_5$), whereas $D\colon \mathcal{P}_5(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ *is*. ◁

Example 3.22 (Computing an image). The image of the map $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ of Example 3.16 is the column space of the matrix A. The columns of A are $(1,2)$, $(2,4)$, $(-1,1)$. But $(2,4) = 2(1,2)$. So only two linearly independent columns remain, and

$$\operatorname{im} T = \operatorname{span}\big((1,2),(-1,1)\big) = \mathbb{R}^2, \qquad \operatorname{rank} T = 2.$$

Hence T is surjective, and the dimensions balance: $3 = \dim \ker T + \operatorname{rank} T = 1 + 2$. This equation is a concrete instance of the theorem of the next section. ◁

A linear map carries spanning lists to spanning lists, and—when injective—it preserves independence.

Proposition 3.23 (Action on lists). Let $T \in \mathcal{L}(V,W)$ and $v_1,\dots,v_n \in V$.

1. If $v_1,\dots,v_n$ spans V, then $Tv_1,\dots,Tv_n$ spans $\operatorname{im} T$.
2. If T is injective and $v_1,\dots,v_n$ is linearly independent, then $Tv_1,\dots,Tv_n$ is linearly independent.

Proof. (1) Every element of $\operatorname{im} T$ is Tv for some $v = \sum_k a_k v_k$, and then $Tv = \sum_k a_k\, Tv_k$ lies in $\operatorname{span}(Tv_1,\dots,Tv_n)$. (2) If $\sum_k c_k\, Tv_k = 0$ then $T\big(\sum_k c_k v_k\big) = 0$, so $\sum_k c_k v_k$ is in $\ker T = \{0\}$ by injectivity; independence of the v_k then forces every $c_k = 0$. ∎

6 The Fundamental Theorem of Linear Maps

The kernel records what T destroys; the image records what it builds. The next theorem—the central result of the lecture—says their dimensions exactly account for the dimension of the domain (Figure 1).

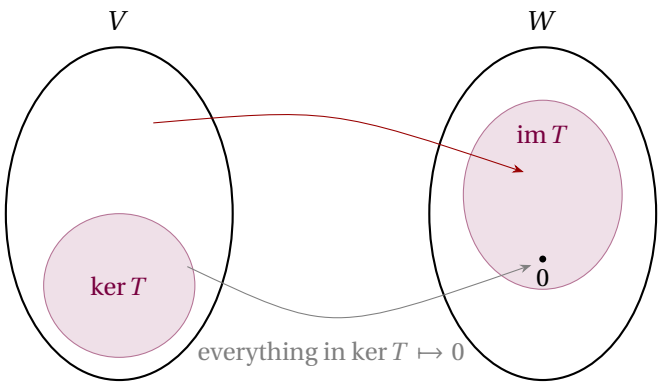


Figure 1: A linear map collapses its kernel to 0 and carries a complement of the kernel one-to-one onto its image; the dimensions balance as $\dim V = \dim \ker T + \dim \operatorname{im} T$.

Theorem 3.24 (Rank–nullity). Suppose V is finite-dimensional and $T \in \mathcal{L}(V,W)$. Then $\operatorname{im} T$ is finite-dimensional and

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Proof. Let $u_1,\dots,u_m$ be a basis of $\ker T$, so $\dim \ker T = m$. Extend it (Lecture 2) to a basis $u_1,\dots,u_m,v_1,\dots,v_n$ of V, so $\dim V = m+n$. We show $Tv_1,\dots,Tv_n$ is a basis of $\operatorname{im} T$, which gives $\dim \operatorname{im} T = n$ and finishes the count.

Spanning. Any $v \in V$ is $v = \sum_i a_i u_i + \sum_j b_j v_j$; applying T and using $Tu_i = 0$ leaves $Tv = \sum_j b_j\, Tv_j$. Hence $Tv_1,\dots,Tv_n$ spans $\operatorname{im} T$ (and $\operatorname{im} T$ is finite-dimensional).

Independence. Suppose $\sum_j c_j\, Tv_j = 0$. Then $T\left(\sum_j c_j v_j\right) = 0$, so $\sum_j c_j v_j \in \ker T$ and can be written $\sum_i d_i u_i$. Thus

$$\sum_j c_j v_j - \sum_i d_i u_i = 0.$$

Independence of the basis $u_1, \dots, u_m, v_1, \dots, v_n$ forces all $c_j = 0$ (and all $d_i = 0$). So $Tv_1, \dots, Tv_n$ is independent, hence a basis of $\operatorname{im} T$. ∎

Example 3.25 (Projection in $\mathbb{R}^2$). Let $P \in \mathcal{L}(\mathbb{R}^2)$ be the projection onto the x-axis, $P(x, y) = (x, 0)$ (Figure 2). The kernel of the projection P is the y-axis, all of which is sent to the zero vector, and its image is the x-axis. So $\dim \ker P = 1$ and $\dim \operatorname{im} P = 1$, and $2 = 1+1 = \dim \mathbb{R}^2$. ◁

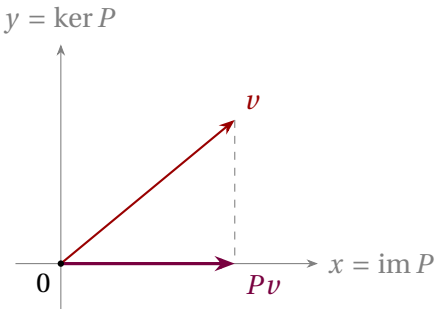


Figure 2: Projection onto the x-axis: the kernel is the y-axis, the image is the x-axis, and $\dim \ker P + \dim \operatorname{im} P = 1 + 1 = 2$.

Comparing the dimensions of the domain and the codomain gives two corollaries.

Corollary 3.26 (Dimension obstructions). Let V, W be finite-dimensional and $T \in \mathcal{L}(V, W)$.

1. If $\dim V > \dim W$, then T is not injective.
2. If $\dim V < \dim W$, then T is not surjective.

Proof. (1) $\dim \ker T = \dim V - \dim \operatorname{im} T \geq \dim V - \dim W > 0$, so $\ker T \neq \{0\}$ and T is not injective by Theorem 3.18. (2) $\dim \operatorname{im} T = \dim V - \dim \ker T \leq \dim V < \dim W$, so $\operatorname{im} T \neq W$. ∎

Example 3.27 (Deciding injectivity and surjectivity). Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be

$$T(z_1, z_2, z_3, z_4) = (z_1 + z_4,\ z_2 + z_3,\ z_1 + z_2).$$

Since $\dim \mathbb{R}^4 > \dim \mathbb{R}^3$, Corollary 3.26(1) forbids injectivity—no computation is needed.

Swapping the spaces gives the second case: no linear map $S\colon \mathbb{R}^3 \to \mathbb{R}^4$ is surjective, since $\dim \mathbb{R}^3 < \dim \mathbb{R}^4$. Hence for some right-hand sides $b \in \mathbb{R}^4$ the system $Sx = b$ has no solutions. Comparing dimensions alone settles both questions. ◁

Corollary 3.28 (Operators: injective = surjective). If V is finite-dimensional and $T \in \mathcal{L}(V)$, then T is injective if and only if it is surjective (and then bijective).

Proof. By Theorem 3.24, $\dim \ker T + \dim \operatorname{im} T = \dim V$. Then

$$\ker T = \{0\} \iff \dim \operatorname{im} T = \dim V \iff \operatorname{im} T = V.$$

Here we use Theorem 3.18 and the fact that a subspace of full dimension is the whole space. ∎

Corollary 3.28 fails in infinite dimensions: the backward shift on $\mathbb{F}^\infty$ is surjective but not injective, while the forward shift $(x_1, x_2, \dots) \mapsto (0, x_1, x_2, \dots)$ is injective but not surjective. This contrast is a first warning of the difficulties that await in Hilbert space (Lectures 13–14).

Example 3.29 (A rotation is an invertible operator). Rotation of $\mathbb{R}^2$ by an angle θ,

$$R_\theta(x, y) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta),$$

is linear, and its kernel is $\{0\}$: a rotation carries every nonzero vector to a nonzero vector. By Corollary 3.28 the rotation is therefore also surjective, hence invertible. Its inverse is $R_{-\theta}$, as geometry demands. This is why symmetry operations in physics are invertible linear operators. ◁

Example 3.30 (Solving an equation by counting dimensions). For every polynomial $q \in \mathcal{P}_m(\mathbb{R})$, is there a p such that $\big((t^2+1)p\big)'' = q$? Define $T : \mathcal{P}_m(\mathbb{R}) \to \mathcal{P}_m(\mathbb{R})$ by

$$Tp = \big((t^2+1)p\big)''.$$

Multiplying by t^2+1 raises the degree by two, and two derivatives lower it by the same amount, so T does map the space $\mathcal{P}_m(\mathbb{R})$ into itself.

Let us find the kernel. If $Tp = 0$, the second derivative of the polynomial $(t^2+1)p$ is zero. Only polynomials of degree at most one, i.e. of the form $at + b$, have zero second derivative. But for nonzero p the polynomial $(t^2+1)p$ has degree at least 2. Only $p = 0$ remains, so $\ker T = \{0\}$ and T is injective.

By Corollary 3.28, T is then also surjective, so the required p exists for *every* q. Injectivity proved existence, although we never constructed the solution itself. ◁

Linear systems, revisited

Rank–nullity is the structural statement behind the row reduction you already know. Take a matrix $A \in M_{m\times n}(\mathbb{F})$ and the map $T(x) = Ax$. The rank of T is the number of pivots, and the nullity is the number of free variables, so

$$(\#\text{free variables}) = n - (\#\text{pivots}) = \dim V - \operatorname{rank} T.$$

The system $Ax = b$ is consistent exactly when $b \in \operatorname{im} T$, i.e. when b lies in the column space. Its full solution set is then the kernel $\ker T$ shifted by a single particular solution x_p.

The same equation explains a long-known fact. If there are more unknowns than equations, then $\dim V > \dim W$, and by Corollary 3.26(1) the map T is not injective. Hence the homogeneous system $Ax = 0$ then always has a nonzero solution.

Example 3.31 (A homogeneous system with free variables). The system

$$x_1 + x_2 - x_3 + 2x_4 = 0, \qquad 2x_1 + 2x_2 + x_3 + x_4 = 0$$

has four unknowns and two equations, so it must have nonzero solutions. We write the coefficients into the rows of a matrix and reduce it (there is no need to write the right-hand side, since it stays zero throughout):

$$\begin{bmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & 1 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 1 & 0 & 1 \\ 0 & 0 & \textcircled{1} & -1 \end{bmatrix} \Longrightarrow\ x_1 = -x_2 - x_4, \quad x_3 = x_4.$$

The pivots are in the first and third columns, so the free variables are x_2 and x_4. Setting $x_2 = 1,\ x_4 = 0$ reads off the vector $(-1, 1, 0, 0)$; setting $x_2 = 0,\ x_4 = 1$ reads off $(-1, 0, 1, 1)$. In

the free-variable positions they carry $(1,0)$ and $(0,1)$, so neither is a multiple of the other. The solution space is

$$\operatorname{span}\big((-1,1,0,0),\ (-1,0,1,1)\big), \qquad \text{of dimension } 2 = 4-2.$$

◁

Example 3.32 (Rank, nullity, and a solution set). Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be given by $Tx = Ax$ with

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{pmatrix}.$$

We reduce the matrix A:

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 2 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Longrightarrow\ x_1 = -2x_2 - x_4, \quad x_3 = -x_4.$$

The pivots are in the first and third columns, so $\operatorname{rank} T = 2$ and, by Theorem 3.24, $\dim\ker T = 4-2 = 2$. The free variables are x_2 and x_4. Setting $x_2 = 1,\ x_4 = 0$ reads off the vector $(-2,1,0,0)$; setting $x_2 = 0,\ x_4 = 1$ reads off $(-1,0,-1,1)$. So

$$\ker T = \operatorname{span}\big((-2,1,0,0),\ (-1,0,-1,1)\big).$$

The image is the column space $\operatorname{span}\big((1,2,1),(0,1,1)\big)$, also of dimension 2, so $4 = 2+2$. Finally $b = (1,3,2)$ is the fourth column Ae_4, so $x_p = (0,0,0,1)$ solves $Ax = b$ and the complete solution set is $x_p + \ker T$. ◁

Example 3.33 (Rank–nullity without matrices). Let D be differentiation on $\mathcal{P}_3(\mathbb{R})$, which has dimension 4. The kernel of differentiation is the constants, so $\dim\ker D = 1$. The image is all derivatives of cubic polynomials, i.e. $\mathcal{P}_2(\mathbb{R})$, of dimension 3. The theorem checks out: $4 = 1+3$. As an operator on $\mathcal{P}_3(\mathbb{R})$ the same D is neither injective nor surjective, but as a map $\mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ it is surjective. Rank and nullity are intrinsic to the map, whereas injectivity and surjectivity depend on the chosen domain and codomain. ◁

Remark 3.34 (Operators in quantum mechanics). Every transformation a physicist applies to a quantum state—a symmetry, a measured observable, the time-evolution operator $e^{-i\hat{H}t/\hbar}$—is linear, precisely so that superpositions transform coherently:

$$T(a|\psi\rangle + b|\phi\rangle) = a\,T|\psi\rangle + b\,T|\phi\rangle.$$

Kernels and images recur throughout: the kernel of the operator $\hat{H} - E\,I$ is the eigenspace at energy E. This is the object the spectral theorem (Lecture 10) will put at centre stage.

7 Measuring the position of a particle

We close the lecture with one question from mechanics, and answering it needs the lecture's main results. A particle moves along a straight line, and a detector records its position $p(t)$ at the three times $t = 0$, $t = 1$ and $t = 2$:

$$p(0) = 3, \qquad p(1) = 6, \qquad p(2) = 13.$$

We want the trajectory p that produced them.

First we must say what a trajectory is allowed to be. Under constant acceleration the position is $p(t) = p(0) + vt + \frac{1}{2}at^2$, a polynomial of degree at most two; if the rate of change j of the acceleration, the *jerk*, is constant as well, a term $\frac{1}{6}jt^3$ is added. The two models are therefore the spaces $\mathcal{P}_2(\mathbb{R})$ and $\mathcal{P}_3(\mathbb{R})$, of dimension 3 and 4. We examine the larger model first.

The measurement map. The three readings are the values of the polynomial p at three fixed points, and the evaluation functional is linear (Example 3.2). The measurements are therefore recorded by a linear map

$$T\colon \mathcal{P}_3(\mathbb{R}) \to \mathbb{R}^3, \qquad Tp = \big(p(0),\, p(1),\, p(2)\big).$$

By Theorem 3.5 the map T is determined by its values on the basis $1, t, t^2, t^3$ of $\mathcal{P}_3(\mathbb{R})$:

$$T1 = (1,1,1), \quad Tt = (0,1,2), \quad Tt^2 = (0,1,4), \quad Tt^3 = (0,1,8).$$

Is the map T injective? Since $\dim \mathcal{P}_3(\mathbb{R}) = 4 > 3 = \dim \mathbb{R}^3$, Corollary 3.26(1) says that it is not. Different trajectories can give the same three readings. The kernel says which ones.

The kernel. Suppose the trajectories p and q give the same readings. Then $Tp = Tq$, so $T(p-q) = 0$ and the difference $p - q$ lies in $\ker T$. The kernel is exactly the ambiguity left in the data. For $p = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ linearity gives $Tp = a_0\, T1 + a_1\, Tt + a_2\, Tt^2 + a_3\, Tt^3$, so we write the values on the basis into the columns of a matrix and reduce it (there is no need to write the right-hand side, since it stays zero throughout):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 0 \\ 0 & \textcircled{1} & 0 & -2 \\ 0 & 0 & \textcircled{1} & 3 \end{bmatrix} \implies a_0 = 0, \quad a_1 = 2a_3, \quad a_2 = -3a_3.$$

The pivots are in the first, second and third columns, so the only free variable is a_3. Setting $a_3 = 1$ reads off the coefficient vector $(0, 2, -3, 1)$, that is, the polynomial $2t - 3t^2 + t^3 = t(t-1)(t-2)$. Hence

$$\ker T = \operatorname{span}\big(t(t-1)(t-2)\big), \qquad \dim \ker T = 1.$$

The generator of the kernel is the cubic polynomial whose roots are exactly the three measurement times. Adding a nonzero multiple of it to a trajectory changes the trajectory at every time except those three, and those three are the times at which the detector reads. None of the three readings can detect the change.

The image and the rank. Is every triple of readings produced by some trajectory? In other words, is $\operatorname{im} T = \mathbb{R}^3$? Rank–nullity answers at once:

$$\operatorname{rank} T = \dim \mathcal{P}_3(\mathbb{R}) - \dim \ker T = 4 - 1 = 3,$$

and a three-dimensional subspace of $\mathbb{R}^3$ is all of $\mathbb{R}^3$. So the map T is surjective: any three readings whatever are produced by some cubic trajectory. Three position measurements can therefore never contradict the cubic model—the model fits every possible outcome.

Recovering the trajectory. Now we use the recorded readings. Which trajectories give them? We solve the equation $Tp = (3, 6, 13)$. We write the values on the basis into the columns of a matrix and the right-hand side after the bar, and reduce the augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 1 & 1 & 1 & 1 & 6 \\ 1 & 2 & 4 & 8 & 13 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} \textcircled{1} & 0 & 0 & 0 & 3 \\ 0 & \textcircled{1} & 0 & -2 & 1 \\ 0 & 0 & \textcircled{1} & 3 & 2 \end{array}\right]$$

$$\implies a_0 = 3, \quad a_1 = 1 + 2a_3, \quad a_2 = 2 - 3a_3.$$

The left-hand block is the matrix of the kernel computation; only the last column is new. Setting $a_3 = 0$ gives the particular solution $p_0(t) = 3 + t + 2t^2$, and indeed $p_0(0) = 3$, $p_0(1) = 6$

and $p_0(2) = 13$. The kernel step already showed that trajectories with the same readings differ by an element of $\ker T$, and conversely $T(p_0 + k) = Tp_0$ for every $k \in \ker T$. The complete solution set is therefore $p_0 + \ker T$:

$$p(t) = 3 + t + 2t^2 + c\,t(t-1)(t-2), \qquad c \in \mathbb{R}.$$

Three position measurements do not determine a cubic trajectory—they determine it only up to an element of the kernel. Figure 3 shows three members of the family.

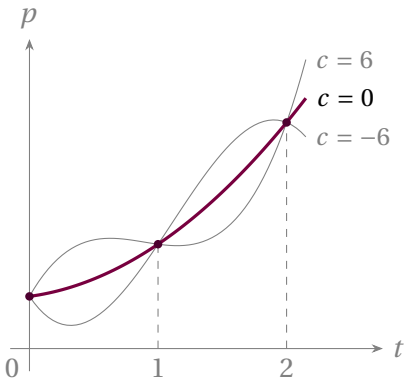


Figure 3: Three trajectories through the same three measured positions (dots): the members $p(t) = p_0(t) + c\,t(t-1)(t-2)$ of the solution family for $c = 0$, which is the particular solution $p_0(t) = 3 + t + 2t^2$, and for $c = 6$ and $c = -6$.

A constant-acceleration trajectory. Take the smaller model instead: the acceleration is constant, so the trajectory is a polynomial of degree at most two. The same formula records the measurements on the smaller space,

$$S\colon \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3, \qquad Sp = \big(p(0),\, p(1),\, p(2)\big),$$

and the kernel needs no new computation: a polynomial in $\ker S$ lies in $\ker T$ and has degree at most two, and the only multiple of $t(t-1)(t-2)$ of degree at most two is the zero polynomial. So

$$\ker S = \ker T \cap \mathcal{P}_2(\mathbb{R}) = \{0\},$$

and the map S is injective by Theorem 3.18. Rank–nullity then gives $\dim \operatorname{im} S = 3 - 0 = 3 = \dim \mathbb{R}^3$, so the map S is surjective as well. Any three readings are produced by exactly one trajectory of constant acceleration: injectivity proved uniqueness and rank–nullity proved existence, though no solution was constructed. For the readings recorded above that trajectory is the one already found, $p_0(t) = 3 + t + 2t^2$: initial position 3, initial velocity 1, acceleration 4, in the units of the recorded times and positions.

The two models differ in one number, $\dim \mathcal{P}_3(\mathbb{R}) = 4$ against $\dim \mathcal{P}_2(\mathbb{R}) = 3$, and rank–nullity turns that difference into the answer: the rank counts the independent readings the measurements can produce, and the nullity counts the directions in which a trajectory can be changed without changing any reading. When the model has more dimensions than there are measurements, Corollary 3.26(1) makes the kernel nonzero, and then no such set of readings determines a trajectory, however accurate the readings are. Two ways out remain: make the model smaller—the step from $\mathcal{P}_3(\mathbb{R})$ to $\mathcal{P}_2(\mathbb{R})$ taken above—or measure more often. Here a fourth reading at any time $t \neq 0, 1, 2$ suffices: at that time $t(t-1)(t-2)$ is nonzero, the rank goes up, and the cubic trajectory is determined exactly.

Quotient spaces *[self-study]*

The solution set $x_p + \ker T$ of the system $Ax = b$ is not a subspace: when $b \neq 0$, it does not contain the zero vector. Still, its structure is clear—it is a *translate* of the subspace $\ker T$. The notion of a quotient space is the right way to describe such objects.

Definition 3.35 (Quotient space). Let U be a subspace of V. The *coset* of $v \in V$ modulo U is

$$v + U = \{v + u : u \in U\}.$$

Two vectors are *equivalent*, $v \sim v'$, if $v - v' \in U$. The *quotient space* V/U is the set of all cosets, with operations

$$(v + U) + (w + U) = (v + w) + U, \qquad \lambda(v + U) = (\lambda v) + U,$$

making it a vector space.

Both operations are defined through *representatives*, so they need checking. If $v + U = v' + U$ and $w + U = w' + U$, then $v - v' \in U$ and $w - w' \in U$, hence $(v + w) - (v' + w') \in U$ and $\lambda v - \lambda v' = \lambda(v - v') \in U$: the answers $(v + w) + U$ and $(\lambda v) + U$ do not depend on which representatives were chosen. The vector-space axioms are then inherited from V one at a time, with $0 + U = U$ as the zero vector.

Let V be finite-dimensional and let $U \subseteq V$ be a subspace. Extend a basis $u_1, \dots, u_m$ of U to a basis $u_1, \dots, u_m, w_1, \dots, w_k$ of V (the extension theorem of Lecture 2). The cosets $w_1 + U, \dots, w_k + U$ are then a basis of V/U. They span: writing $v = \sum_i \alpha_i u_i + \sum_j \beta_j w_j$ and absorbing the u-part into U gives $v + U = \sum_j \beta_j (w_j + U)$. They are independent: if $\sum_j \beta_j (w_j + U) = 0 + U$ then $\sum_j \beta_j w_j \in U$, so it is also a combination of the u_i, and independence of the full basis forces every $\beta_j = 0$. Counting lengths,

$$\dim(V/U) = \dim V - \dim U.$$

The same statement holds • cited in arbitrary algebraic dimension, in the form $\dim V = \dim U + \dim(V/U)$ as a sum of cardinals. That version is not available here: it needs the general basis-extension theorem, which rests on Zorn's lemma, and a notion of dimension for infinite-dimensional spaces, whereas Lecture 2 proves extension only in a finite-dimensional space and defines dim only there.

The coset $x_p + \ker T$ from the previous subsection is *exactly* an element of $V/\ker T$: the solution set of $Ax = b$ is a single point in the quotient space, not in V itself. Different right-hand sides b correspond to different cosets—the quotient "forgets" what happens inside the kernel.

One word is needed before the next result, ahead of its proper treatment. Two vector spaces are *isomorphic*, written $V \cong W$, if some bijective linear map $V \to W$ exists; such a map is an *isomorphism*. That is all the term means below. Lecture 4 studies it in depth and proves that for finite-dimensional spaces the dimension alone decides it.

Example 3.36 (Cosets in $\mathbb{R}^3$). Let U be the xy-plane:

$$U = \operatorname{span}\big((1,0,0),(0,1,0)\big) = \{(x, y, 0) : x, y \in \mathbb{R}\} \subseteq \mathbb{R}^3.$$

The coset of the vector $v = (a, b, c)$ is the horizontal plane through the point (a, b, c):

$$v + U = \{(a + s,\ b + t,\ c) : s, t \in \mathbb{R}\} = \{(x, y, c) : x, y \in \mathbb{R}\}.$$

This plane is parallel to the plane U and lies at height c (Figure 4). Two vectors have the same coset exactly when their z-coordinates agree. So an element of the quotient space

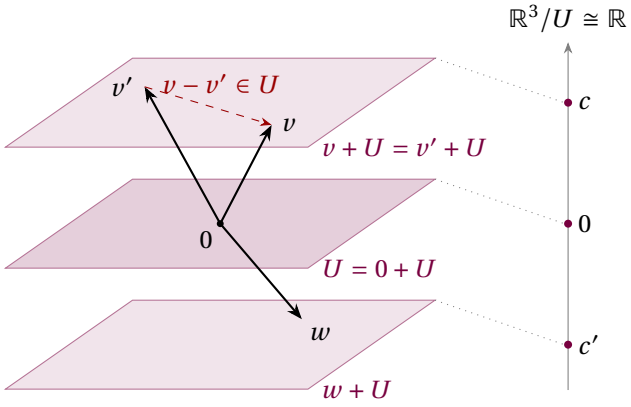


Figure 4: The cosets of the plane U in $\mathbb{R}^3$ are the planes parallel to it. The vectors v and v' lie in the same coset, since $v - v' \in U$. The coset of w is a different plane. Each plane is one element of the quotient space $\mathbb{R}^3/U$, and a single number identifies it: the height of the plane.

$\mathbb{R}^3/U$ is identified by a single number c, and the operations on cosets become operations on these numbers: the sum of the planes at heights c and c' is the plane at height $c + c'$. Hence $\mathbb{R}^3/U \cong \mathbb{R}$ (the z-axis), and the quotient space has dimension $3 - 2 = 1$.

The plane U is the kernel of the map $T(x, y, z) = z$, and the plane at height c is the solution set of the equation $T(x, y, z) = c$. Different right-hand sides c correspond to different cosets. ◁

Theorem 3.37 (First Isomorphism Theorem). For $T \in \mathcal{L}(V, W)$, the map $\tilde{T} : V/\ker T \to W$ defined by

$$\tilde{T}(v + \ker T) = Tv$$

is well-defined, linear, and injective, with $\operatorname{im} \tilde{T} = \operatorname{im} T$. Hence

$$V/\ker T \cong \operatorname{im} T,$$

without any finite-dimensional hypothesis. If V is finite-dimensional, then comparing dimensions in the formula $\dim(V/\ker T) = \dim V - \dim \ker T$ recovers the rank–nullity equation $\dim V = \dim \ker T + \dim \operatorname{im} T$. This is an *alternative view* of Theorem 3.24, already proved directly above, rather than a new, independent proof of it.

Proof. For brevity, write $K = \ker T$. (1) *Well-defined.* The same coset can be written with different representatives. If $v + K = v' + K$, then $v - v' \in K$, so

$$Tv - Tv' = T(v - v') = 0.$$

Hence $Tv = Tv'$, and the value $\tilde{T}(v + K)$ does not depend on the representative. (2) *Linear.* The operations on the quotient act on representatives (Definition 3.35), so

$$\tilde{T}\big((v + K) + (w + K)\big) = \tilde{T}\big((v + w) + K\big) = T(v + w) = Tv + Tw = \tilde{T}(v + K) + \tilde{T}(w + K)$$

and

$$\tilde{T}\big(\lambda(v + K)\big) = \tilde{T}\big((\lambda v) + K\big) = T(\lambda v) = \lambda\, Tv = \lambda\, \tilde{T}(v + K).$$

(3) *Injective.* If $\tilde{T}(v + K) = 0$, then $Tv = 0$, i.e. $v \in K$. So $v + K = 0 + K$ is the zero vector of V/K. Hence $\ker \tilde{T} = \{0 + K\}$, and $\tilde{T}$ is injective by Theorem 3.18. (4) *Image.* The values of $\tilde{T}$

are $\tilde{T}(v+K) = Tv$ as v runs over all of V. Therefore

$$\operatorname{im}\tilde{T} = \{Tv : v \in V\} = \operatorname{im} T.$$

So the map $\tilde{T}\colon V/K \to \operatorname{im} T$ is linear, injective and surjective, i.e. an isomorphism, and $V/\ker T \cong \operatorname{im} T$. (5) *Dimensions.* Let V be finite-dimensional. Above we found a basis $w_1+K,\dots,w_k+K$ of V/K, where $k = \dim V - \dim K$. The map $\tilde{T}$ sends this basis to the vectors $Tw_1,\dots,Tw_k$. By Proposition 3.23 they span $\operatorname{im}\tilde{T} = \operatorname{im} T$ and, since $\tilde{T}$ is injective, they are linearly independent. So they form a basis of $\operatorname{im} T$ and $\dim\operatorname{im} T = k$. Therefore

$$\dim V = \dim\ker T + k = \dim\ker T + \dim\operatorname{im} T. \qquad \blacksquare$$

The First Isomorphism Theorem is the canonical form of rank–nullity: it does not merely count dimensions but exhibits an explicit isomorphism.

Remark 3.38 (Physics: gauge fields *[self-study]*). Quotient spaces arise naturally in modern physics. For example, in electromagnetism the four-potential A_μ is not unique: two potentials related by the transformation $A_\mu \to A_\mu + \partial_\mu\chi$ produce the same field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The "pure gauge" configurations (those of the form $\partial_\mu\chi$) form a subspace of the space of all potentials, and the space of fields that physics "sees" as distinct is the quotient. A gauge-fixed description chooses one representative from each coset—just as choosing x_p picks one solution from each coset $x_p + \ker T$. You will meet gauge fields in the Electrodynamics course.

Summary of main concepts

A linear map preserves linear combinations, hence superposition. It is fixed by its values on a basis, which may be chosen freely (Theorem 3.5). Its kernel $\ker T$ and image $\operatorname{im} T$ are subspaces; T is injective exactly when $\ker T = \{0\}$ and surjective exactly when $\operatorname{im} T = W$. The rank–nullity theorem $\dim V = \dim\ker T + \dim\operatorname{im} T$ ties the two together, yielding at a glance that maps to smaller spaces cannot be injective, maps to larger spaces cannot be surjective, and operators on a finite-dimensional space are injective if and only if surjective. In coordinates this is the bookkeeping of pivots and free variables behind every linear system.

- **Linear map**—preserves linear combinations (superposition); fixed by its values on a basis.
- **Kernel and image**—subspaces $\ker T$ and $\operatorname{im} T$; injective $\iff \ker T = \{0\}$, surjective $\iff \operatorname{im} T = W$.
- **Rank–nullity**—$\dim V = \dim\ker T + \dim\operatorname{im} T$.
- **Operators: injective = surjective**—on a finite-dimensional space, injective $\iff$ surjective.
- **Quotient space; First Isomorphism Theorem**—$V/\ker T \cong \operatorname{im} T$ (self-study); gauge equivalence in physics.